
Vera: A Layered Diffusion Model for Content-Preserving Video Editing

Hongkai Zheng^{1,2†} Ta-Ying Cheng² Benjamin Klein²
Yisong Yue¹ Zhuoning Yuan^{2‡}

¹California Institute of Technology ²Netflix, Inc.

<https://vera-layered-diffusion.github.io/>

Abstract

Video diffusion models have enabled remarkable progress in video generation and editing. However, content preservation remains a core challenge: existing methods regenerate every pixel and often alter elements that should remain unchanged, such as characters or background scenes. We introduce Vera, a layered diffusion framework for content-preserving video editing. Instead of regenerating the entire video, Vera generates an edit layer along with an alpha matte for compositing with the source video, separating creative editing from content preservation by design. To ensure coherent edits that blend seamlessly with the original content, we extend the text-to-video DiT into a Mixture-of-Transformers (MoT) architecture, with separate DiTs for each layer that interact through joint self-attention. To support the training of Vera, we further construct a high-quality layered dataset with accurate alpha mattes, diverse scenes and dynamics, and visual effects. Across both our quantitative benchmark and a human preference study, Vera outperforms state-of-the-art video editing models in content preservation without sacrificing edit quality—despite training on only 486K frames of layered data, which is orders of magnitude fewer than existing methods.

1 Introduction

Video generation has recently achieved remarkable breakthroughs with cinematic-level quality (Google, 2026; OpenAI, 2025). Driven by advancements in text-to-video (T2V) foundation diffusion models (Wan et al., 2025b; Yang et al., 2024; Kong et al., 2024; HaCohen et al., 2024), controllable video editing has attracted significant attention for narrowing the gap between video generation and practical production use. Recent progress spans object insertion and removal, background replacement, visual effects (VFX), style transfer, and relighting (Bian et al., 2025b; Tang et al., 2023; Zhang et al., 2025; Litman et al., 2026; Fu et al., 2025; Li et al., 2025; Zhou et al., 2025). These capabilities streamline complex manual editing tasks that traditionally take weeks, and lower the barrier for creators to focus on creativity rather than technical execution. Despite the progress, a key challenge remains **content-preserving editing**: regions outside the target edited area are often inadvertently modified. In commercial production, even a small unintended change can render an edit unusable and erode user trust in the tool.

Existing approaches attempt to address this through various strategies such as regional constraints and mask conditioning (Zhang et al., 2025; Jiang et al., 2025; Litman et al., 2026; Jia et al., 2025), aiming to isolate targeted edits from unrelated regions. However, these methods still operate within the end-to-end diffusion paradigm, which has two fundamental limitations: (i) even with explicit region control, the model regenerates the entire video and can still introduce inadvertent changes to regions that should be strictly preserved, especially in complex scenarios; (ii) the end-to-end paradigm produces only a final composite, whereas practical production workflows often require layered assets for iterative editing—manually separating added content from the output for subsequent edits is tedious and error-prone.

Layered approach (Yin et al., 2025; Ji et al., 2025; Wang et al., 2025; Dong et al., 2025b) offers a principled alternative by producing edit content as a separate layer with an alpha matte for compositing, preserving the

† Work done during an internship at Netflix.

‡ Project Lead. Correspondence: Zhuoning Yuan <zyuan@netflix.com>.

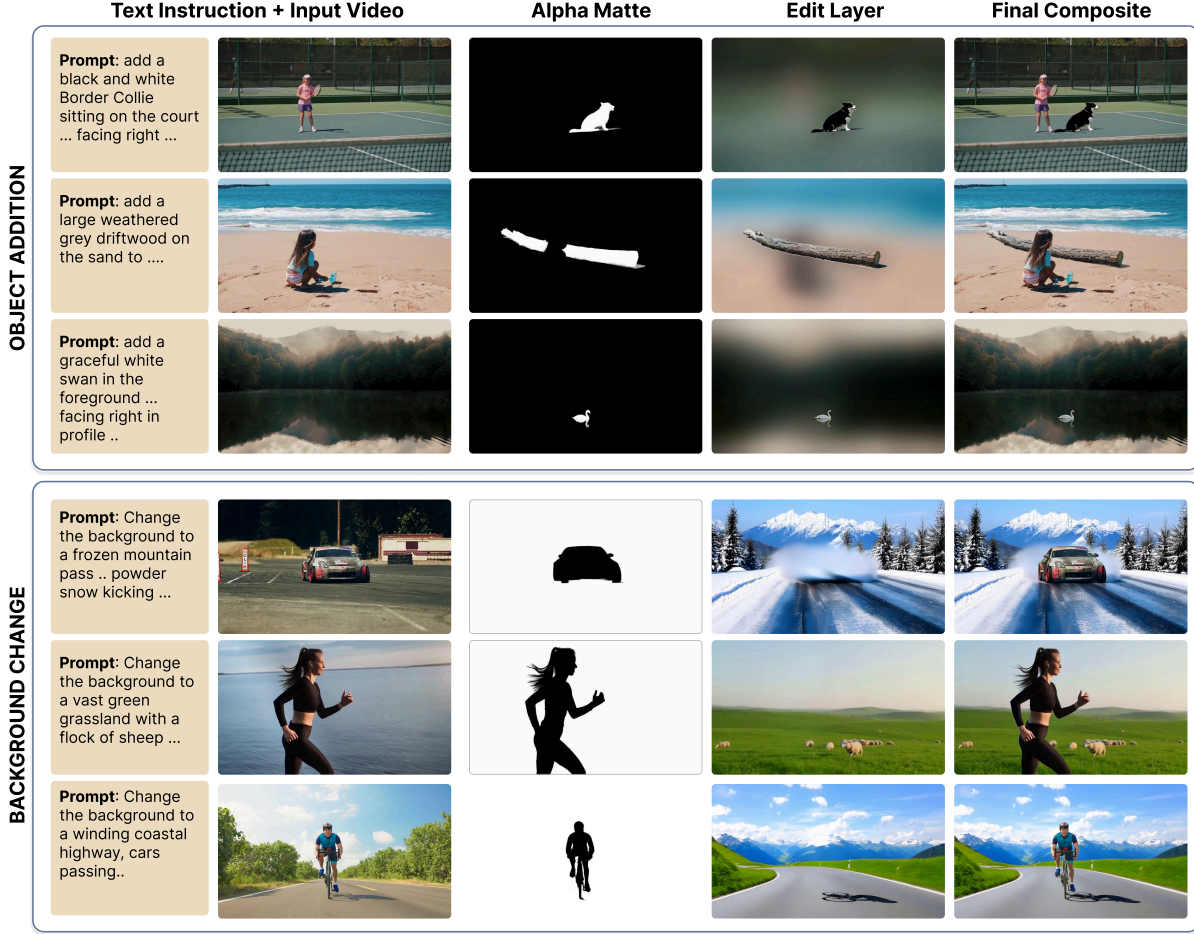


Figure 1. Given an input video and a text instruction, Vera generates an edit layer together with an alpha matte that can be directly composited with the input video to produce the edited result. For object addition, the alpha includes the effects (e.g. shadows) to be added into the composite; for background replacement, Vera learns to include the effects that complements the preserved regions (e.g. smoke behind cars). The joint generation of edit layer and its matte allows the original content to be preserved during editing, even for very fine-grained details (e.g. hair of the girl during background change). The input prompts in the figure have been shortened for visual clarity.

original video by construction. However, existing work in this direction has focused primarily on text-to-RGBA video generation rather than editing, and applying layered generation to video editing introduces significant new challenges and largely remains underexplored. The edit layer and alpha matte must be precisely aligned, and the generated content must be highly consistent with the source video—matching camera motion, lighting, spatial layout, and scale—to achieve natural compositing. The model must also handle complex cross-layer interactions such as shadows, reflections, and occlusions. These challenges demand both an architecture capable of coherent cross-layer generation and high-quality layered training data that captures such interactions.

We introduce **Vera** (from the Latin *vēra*, meaning "genuine"), a layered diffusion framework that addresses these challenges for content-preserving video editing. As depicted in Fig. 2, given a source video and an editing instruction, Vera generates three outputs: an edit layer, an alpha matte, and a composite video. The edit layer and alpha matte composite with the source video to produce the final result, explicitly separating the generated edit from the original content. To ensure coherent edits that blend seamlessly with the source video, we extend the text-to-video DiT into a MoT architecture with separate DiTs per layer that interact through joint self-attention, and curate a high-quality layered dataset with accurate alpha mattes, diverse dynamics, and visual effects. Our contributions are as follows:

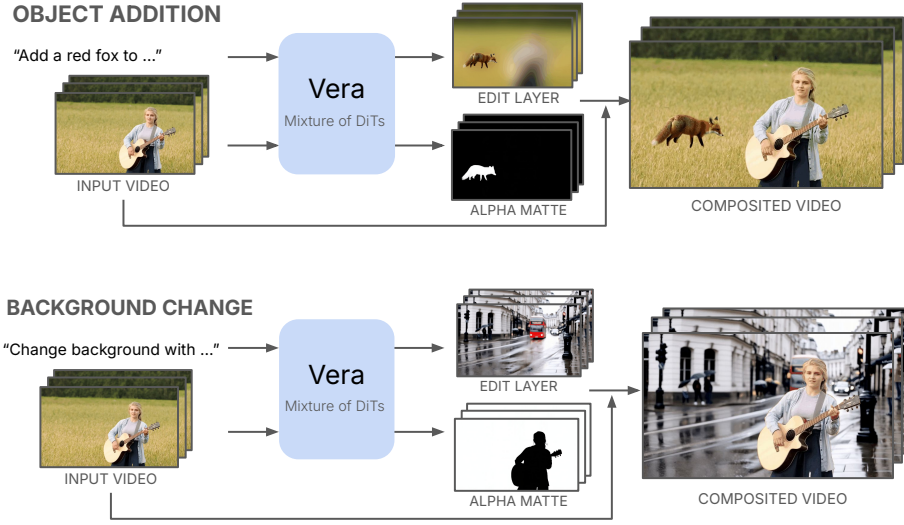


Figure 2. Overview of the Vera inference pipeline. Given an input video and a text editing instruction, Vera’s MoT architecture jointly generates an edit layer, an alpha matte, and a composite video. The edit layer and alpha matte are then composited with the source video to produce the final edited output.

- We propose Vera, a new layered video editing framework that preserves content integrity by generating edits as a separate RGBA layer, enabling both faithful preservation and natural composition.
- We construct a high-quality layered video dataset comprising synthetic composites, realistic single- and multi-object scenes, and scenes with interactive visual effects, along with a test benchmark spanning diverse motion dynamics and scene complexity.
- Measured by both standard quantitative metrics and a human preference study, Vera outperforms state-of-the-art video editing models in content preservation without sacrificing edit quality—despite training on 486K frames, orders of magnitude fewer than existing methods.
- Through controlled experiments, we identify the key architecture and data choices that enable a strong layered diffusion model with both faithful preservation and high-quality composition.

2 Related Work

2.1 Diffusion Models for Video Generation and Editing

The advent of diffusion models (Ho et al., 2020; Liu et al., 2022) transformed the landscape of generative vision models, revolutionizing high-fidelity image synthesis (Esser et al., 2024; Batifol et al., 2025) and soon video generation (Google, 2026; Esser et al., 2023; Wan et al., 2025a). Recently, proprietary models such as Runway’s Gen series (Esser et al., 2023; Runway, 2025) and Google’s Veo 3 (Google, 2026) have shown powerful capabilities, offering highly controllable text-to-video generation and editing. Concurrently, several open-source video models such as LTX-Video (HaCohen et al., 2024), CogVideo (Yang et al., 2024), and WAN (Wan et al., 2025a), have also shown promising capabilities in generating temporally consistent videos.

Adapting these powerful generative priors for video editing has also become a highly active area of research. Several approaches now enable temporally consistent, prompt-driven video editing (Jiang et al., 2025; Geyer et al.). Furthermore, the scope of editing has expanded toward end-to-end visual effects (VFX) generation (Li et al., 2025; Bian et al., 2025a), allowing for complex structural and stylistic scene modifications. To support the training and evaluation of these complex editing frameworks, comprehensive video editing datasets have also been recently introduced to the community (Zi et al.; Hu et al., 2024; Zhang et al., 2025). However, one major drawback of these models is the faithfulness of unchanged regions – the nature of denoising a new video probabilistically leads to many unintended changes in places not imposed by the text

prompt. Vera aims to mitigate this issue by proposing an end-to-end layered framework that generates both the edited video and its corresponding alpha matte.

2.2 Layer-wise Image Video Generation

To achieve granular control while maintaining faithfulness over scene composition, some recent work has gravitated toward layer-wise image and video generation strategies. LayerDiffuse (Zhang & Agrawala, 2024) is one of the first to generate transparent images in multiple transparent layers. LayerFusion (Dalva et al., 2024) generates images with two layers jointly – a foreground RGBA layer with a background layer – to enhance the harmonization. LayerDecomp (Yang et al., 2025) decomposes an image into RGB layer containing effects like shadows for edits. LayerEdit (Fu et al., 2025) proposes a training-free method that decomposes objects into layers through the model’s attention for more disentangled multi-object editing. More recently, Qwen-Image-Layered (Yin et al., 2025) decomposes an image into multiple RGBA layers via a multimodal diffusion transformer.

Other work have extended transparent and layered visual generation into videos. LayerFlow (Ji et al., 2025) is one of the first to propose layer-wise video generation from per-layer prompts. Transpixeler (Wang et al., 2025) builds on top of pretrained video models into additional outputting alpha channel while keeping its original RGB capabilities. Wan-Alpha (Dong et al., 2025a) learns to shift away the alpha distribution from the RGB distribution to enable better RGBA video generation. On the other hand, notable works like Generative Omnimate (Lee et al., 2025) decomposes videos into multiple layers, which can then be used for video editing. These methods, however, lack the high-quality video data for training and struggle with complex interactions like shadows and reflections.

3 Method

We consider the content-preserving video editing problem, where the source video $V_{\text{src}} \in \mathbb{R}^{T \times H \times W \times 3}$ is a composition of two parts:

$$V_{\text{src}} = (1 - A_{\text{edit}}) \circ V_{\text{preserved}} + A_{\text{edit}} \circ V_{\text{non-preserved}}, \quad (1)$$

where $V_{\text{preserved}} \in \mathbb{R}^{T \times H \times W \times 3}$ is the content that should remain untouched, $A_{\text{edit}} \in [0, 1]^{T \times H \times W}$ is an alpha matte, and $\circ$ denotes the Hadamard product. Given the source video V_{src} and a conditioning signal C (e.g., a text prompt and an optional mask), our goal is to generate an edit layer V_{edit} along with the corresponding alpha matte A_{edit} . The output video is then composited as:

$$V_{\text{composite}} = (1 - A_{\text{edit}}) \circ V_{\text{preserved}} + A_{\text{edit}} \circ V_{\text{edit}}. \quad (2)$$

While A_{edit} and $V_{\text{preserved}}$ are largely constrained by the source video and conditioning signal, they are not directly available and must be inferred by the model. In binary-mask regions, $V_{\text{preserved}}$ is fully determined by the mask and the source video. In semi-transparent regions, however, the source video is a mixture of preserved and non-preserved content, and the model must disentangle them. In this work, we focus on the common setting where semi-transparent regions are small so that $V_{\text{preserved}}$ can be well approximated by V_{src} . This layered formulation explicitly separates creative edit generation from content preservation, maintaining the integrity of the original video.

Remark 3.1. The above approximation requires that $V_{\text{preserved}}$ has minimal semi-transparent regions, which holds for common editing applications, including background replacement and object addition. Importantly, the edit layer V_{edit} is fully generative and may contain large semi-transparent regions (e.g., shadows, reflections, or glasses) without affecting the approximation. Our modeling framework and architecture are general and apply equally to cases where $V_{\text{preserved}}$ itself involves significant semi-transparency. The primary bottleneck for handling such cases lies in constructing suitable layer training data, which we leave to future work.

3.1 Vera Framework and Architecture

3.1.1 Modeling framework

By the compositing equation (Eq. 2), any three of the four variables V_{edit} , A_{edit} , $V_{\text{preserved}}$, and $V_{\text{composite}}$ fully determine the fourth. We therefore model the joint conditional distribution $p(V_{\text{edit}}, A_{\text{edit}}, V_{\text{composite}} | V_{\text{src}}, C)$

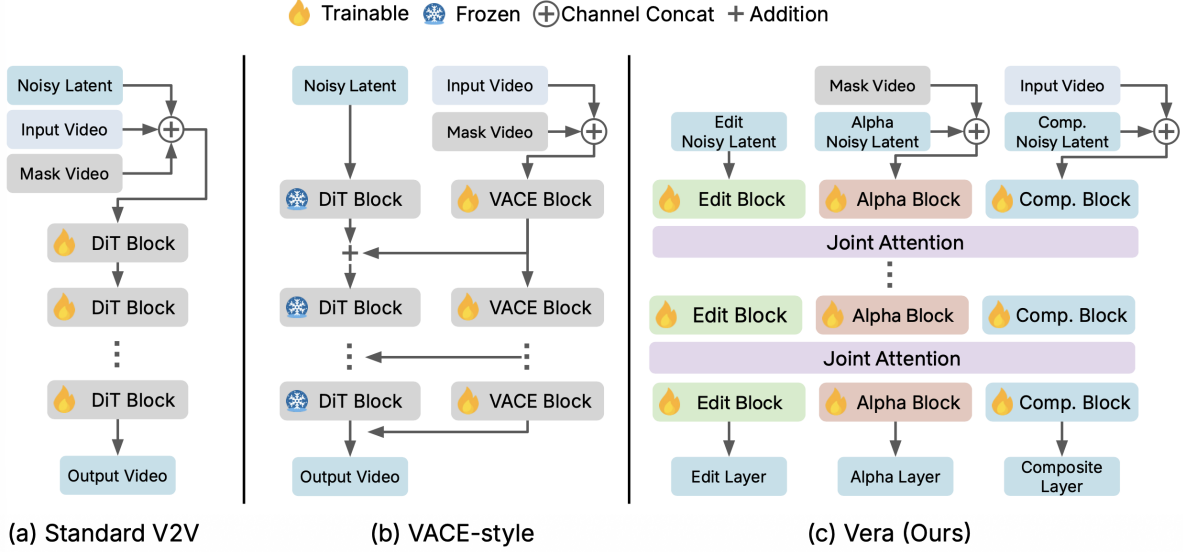


Figure 3. The architecture of Vera compared to other video editing methods. VAE encoding, VAE decoding, and patchifying are omitted for clarity. (a) Standard fine-tuning of a pretrained T2V model for video editing. (b) VACE-style (Jiang et al., 2025) fine-tuning with additional context adapter blocks. (c) Vera consists of three DiTs, each responsible for modeling a separate layer, with interactions across layers enabled through joint self-attention. Ablations on this is presented in Section 4.3.

using a diffusion model (Song et al.). We choose to include $V_{\text{composite}}$ over $V_{\text{preserved}}$ because $V_{\text{composite}}$ shares the distribution of natural videos, which aligns better with the pretrained video generation model. Once these three quantities are generated, the preserved content can be recovered as:

$$V_{\text{preserved}} = \begin{cases} \frac{V_{\text{composite}} - A_{\text{edit}} \circ V_{\text{edit}}}{1 - A_{\text{edit}}}, & \text{in semi-transparent regions,} \\ V_{\text{src}}, & \text{elsewhere.} \end{cases} \quad (3)$$

In our experiments, we use the approximation $V_{\text{preserved}} \approx V_{\text{src}}$. Note that V_{edit} is mathematically undefined in the region where $A_{\text{edit}} = 0$. We set the undefined region to a Gaussian smoothed version of the composite for better edge blending and consistency.

We implement this with a latent diffusion model framework. All videos are encoded into a latent space using the Wan2.1 VAE (Wan et al., 2025b), where the alpha matte sequence is treated as a video with identical RGB channels. We denote the latents of V_{src} , V_{edit} , A_{edit} , and $V_{\text{composite}}$ as Z_{src} , Z_{edit} , Z_{alpha} , and $Z_{\text{composite}}$, respectively. Using the flow matching (Lipman et al.; Liu et al.) parameterization, we train a neural network u_{θ} that takes Z_{src} and C as inputs and jointly predicts the velocity fields for all the layers: $u_{\theta} = [u_{\theta;\text{edit}}, u_{\theta;\text{alpha}}, u_{\theta;\text{composite}}]$. The training objective is:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, C, Z_{\text{src}}, Z_1, Z_0} \|u_{\theta}(Z_t; t, C, Z_{\text{src}}) - (Z_1 - Z_0)\|_2^2, \quad (4)$$

where $Z_0 \sim \mathcal{N}(0, I)$, $Z_1 = [Z_{\text{edit}}, Z_{\text{alpha}}, Z_{\text{composite}}]$, $t \in [0, 1]$, and $Z_t = tZ_1 + (1 - t)Z_0$.

3.1.2 Architecture design

A key design challenge is that the three generated quantities (edit layer V_{edit} , alpha matte A_{edit} , and composite video $V_{\text{composite}}$) have substantially different distributions: V_{edit} contains decoupled creative content, A_{edit} is a grayscale matte that depends not only on the edit content but also on interactions between the edit and the original scene (e.g., an inserted object partially occluded by a foreground subject), and $V_{\text{composite}}$ is a natural video. A single shared transformer would need to reconcile these distributional differences entirely through training, which we found to be data-inefficient (Team, 2024).

Following the Mixture-of-Transformers (MoT) framework (Liang et al., 2025), we use three separate DiTs—one per output—that interact through joint self-attention as shown in Fig. 3. While MoT was originally

proposed for multi-modal learning, we find it equally effective for modeling the interactions between output layers with distinct distributions. Each DiT maintains its own QKV projections and FFN weights, but tokens from all three DiTs are concatenated into a single sequence for the self-attention operation, enabling cross-layer interaction while allowing each branch to specialize.

All three DiTs are initialized from the same pretrained text-to-video model (Wan et al., 2025b). To incorporate conditioning video inputs, we introduce two additional patch embedding layers: one for the input video and one for an optional mask video. The source video tokens are added to the composite tokens, while the mask tokens are added to the noisy alpha tokens. All layer tokens share the same positional encoding (RoPE). To allow the model to distinguish between layers, we add a zero-initialized learnable embedding to the alpha and composite layer tokens, respectively.

3.2 Data Construction and Curation

Since no public dataset provides layered video editing data suitable for training our model, we construct a layered dataset from open-source videos using a combination of annotation and generation tools. Fig. 4 shows an overview of our layered training data. The entire dataset contains 486K frames in 832×480 resolution, categorized into complementary subsets of increasing complexity (see pie chart in Fig. 4). We give a brief overview of each subset below; the data pipelines are illustrated in Fig. 5 and described in detail in Appendix A.1.1.

Synthetic composites (Object Addition and Background Change). We derive layered foreground-background data from VideoMatte240K (Lin et al., 2021), which provides carefully annotated alpha mattes with precise alpha values for fine structures such as hair – details that are difficult to obtain from automated matting tools. Since VideoMatte240K contains only foreground mattes without backgrounds, we generate diverse synthetic backgrounds using an inpainting model (Jiang et al., 2025) and composite them with the extracted foregrounds. This subset provides accurate alpha supervision for both background change and object addition, but is limited to human subjects filmed with centered, static cameras.

Realistic single-object videos (Object Addition and Background Change). To introduce natural scene diversity and camera motion, we source real-world videos from Pexels (pex) and Mixkit (mix). We build a multi-stage data pipeline that generates layered data via a chain of segmentation (Ravi et al.), video matting (Lim et al., 2026), video inpainting and generation (Jiang et al., 2025; Zi et al., 2025), and human annotation and filtering. These videos feature diverse scenes and dynamic camera motion, but typically contain a single prominent subject with limited visual effects. This subset provides training data for both background change and object addition.

Realistic multi-object videos with effects (Object Addition only). To handle more complex scenes, we extend the pipeline above with an additional omnimatte optimization step (Lee et al., 2025) and an alpha matte curation stage, enabling the extraction of individual objects together with their associated visual effects (e.g., shadows and reflections). This subset features multiple objects in complex scenes, rich dynamics, and provides data exclusively for the object addition task.

4 Experiments

We evaluate Vera on two representative video editing tasks – background change and object addition – to assess its content preservation, video quality, and instruction compliance. Sec. 4.1 describes the experimental setup, including training details, evaluation protocol, and baselines. Sec. 4.2 presents both qualitative and quantitative comparisons against existing methods, followed by a human preference study in Sec. 4.2.2. Sec. 4.3 ablates architectural design choices and the impact of different training data.

4.1 Experimental Setup

4.1.1 Training

We train two model variants: Vera-1.3B and Vera-14B. Vera-1.3B initializes each of its three DiTs from Wan2.1-1.3B text-to-video model (Wan et al., 2025b), yielding 3.9B parameters in total. Vera-14B initializes each DiT from Wan2.1-14B, yielding 42B parameters in total. We train for 8,000 steps with a batch size of 16 at 832×480 resolution and 49 frames per clip. During training, we randomly drop the mask input so that both models learn to operate with and without mask video input.

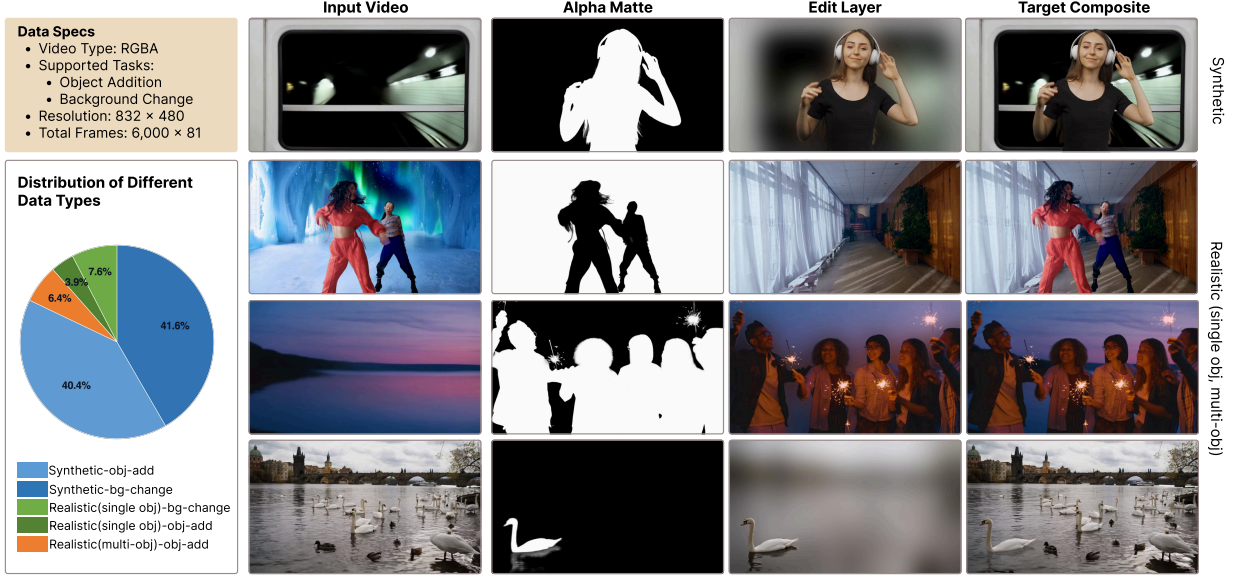


Figure 4. Overview of our layered training data. Each sample consists of an input video, an edit layer, an alpha matte, and a composite target video. The white regions in the edit layer indicate transparency. We curate data for two tasks: background change and object addition, including samples with interactive effects such as shadows and reflections.

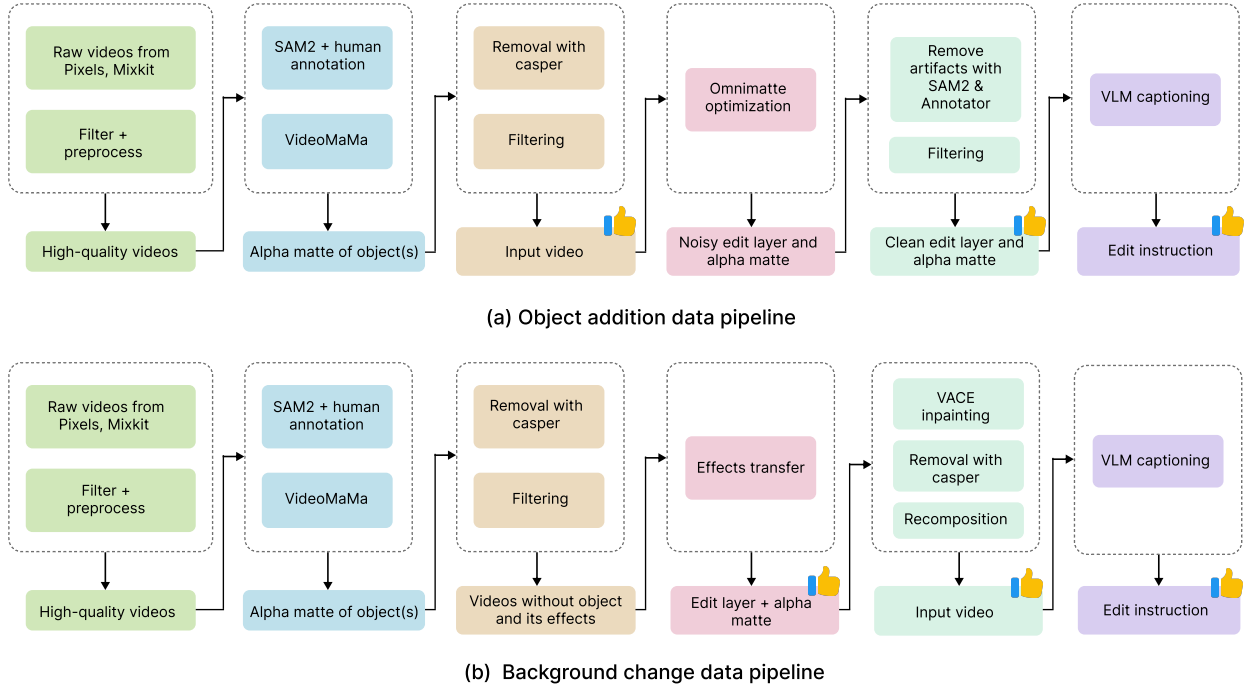


Figure 5. Overview of the data construction pipelines for (a) object addition and (b) background change. Each color represents a distinct stage; dashed-line blocks denote the operations within a stage, and the block outside the dashed line is the stage output. Thumb-up icons denote the final outputs to be used for model training and evaluations.

4.1.2 Evaluation

We curate 72 test video-prompt pairs for object addition and 69 for background change, sourced from Pexels (pex), DAVIS (Pont-Tuset et al., 2017; Caelles et al., 2019), and VACEBench (Jiang et al., 2025). The

Table 1. Comparison with existing video editing methods. VLM-based metrics (CS, CT, IS) are averaged over three VLMs. OC and TF are near-identical across many models and not bolded. Best results in other columns are in **bold**. LayerFlow is included for reference only, as it is limited to 16 frames at 720×480 resolution and thus evaluated under different conditions.

Method	Content Preservation			Video Quality				Instruction Compliance	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	OC $\uparrow$	TF $\uparrow$	CS $\uparrow$	CT $\uparrow$	OSC $\uparrow$	IS $\uparrow$
<i>Object Addition</i>									
Ditto (Bai et al., 2025)	13.1	0.447	0.529	0.98	0.98	2.84	3.13	0.239	3.62
Lucy-Edit (Team, 2025)	19.0	0.798	0.301	0.98	0.98	2.54	2.79	0.238	2.98
ICVE (Liao et al., 2025)	18.0	0.818	0.265	0.98	0.98	2.88	3.12	0.239	3.31
VideoPainter (Bian et al., 2025b)	10.7	0.354	0.669	0.64	0.98	2.54	2.80	0.247	2.87
VACE-1.3B (Jiang et al., 2025)	10.3	0.323	0.667	0.98	0.98	2.55	3.01	0.250	2.99
VACE-14B (Jiang et al., 2025)	10.1	0.325	0.666	0.98	0.98	2.63	3.08	0.248	3.26
ReCo (Zhang et al., 2025)	16.8	0.778	0.280	0.98	0.98	3.18	3.35	0.243	3.77
Vera-1.3B	25.3	0.949	0.078	0.98	0.98	3.49	3.47	0.243	4.05
Vera-14B	26.1	0.950	0.082	0.98	0.98	3.59	3.66	0.244	4.13
<i>Background Change</i>									
Ditto (Bai et al., 2025)	21.7	0.888	0.092	0.97	0.97	3.29	3.25	0.240	4.26
Lucy-Edit (Team, 2025)	22.3	0.908	0.078	0.96	0.96	2.50	2.50	0.235	3.71
ICVE (Liao et al., 2025)	26.3	0.950	0.050	0.96	0.97	2.80	2.97	0.228	3.78
VideoPainter (Bian et al., 2025b)	29.9	0.975	0.032	0.83	0.97	2.29	2.31	0.237	3.64
VACE-1.3B (Jiang et al., 2025)	31.5	0.981	0.024	0.96	0.97	3.74	3.61	0.245	4.48
VACE-14B (Jiang et al., 2025)	31.7	0.983	0.021	0.96	0.96	3.68	3.59	0.243	4.44
ReCo (Zhang et al., 2025)	24.9	0.935	0.061	0.96	0.97	2.45	2.71	0.220	3.05
LayerFlow* (Ji et al., 2025)	23.3	0.886	0.087	0.94	0.95	2.51	2.35	0.243	4.14
Vera-1.3B	35.2	0.993	0.010	0.96	0.97	3.34	3.25	0.243	4.35
Vera-14B	36.2	0.993	0.010	0.96	0.97	3.28	3.21	0.242	4.38

test videos span multiple difficulty levels, including slow to fast motion, various camera motions, single and multiple objects, and both simple and complex scenes. For background change, we supply a coarse object mask video obtained from SAM 2 (Carion et al., 2025) to models that support mask input. For object addition, no input mask is provided; models must determine the placement and extent of the inserted object themselves. We evaluate along three complementary dimensions (Table 7): (1) *Content preservation* measures whether regions outside the edit remain unaltered via pixel-level and perceptual similarity. For background change, similarity is computed over the preserved region annotated with alpha mattes; for object addition, it is computed on the full frame. (2) *Video quality* assesses temporal coherence and per-frame spatial quality. (3) *Instruction compliance* measures whether the edited video faithfully executes the editing prompt. We adopt temporal flickering (TF), overall semantic consistency (OSC), and instruction satisfaction (IS) from VBench (Huang et al., 2024) and IVEBench (Chen et al., 2026), and report overall consistency (OC) as the average of VBench’s subject consistency and background consistency. We observe that classic video quality metrics from prior benchmarks often fail to distinguish between recent models. We therefore introduce two VLM-based video quality metrics—composition spatial quality (CS) and composition temporal quality (CT), which we find to be better aligned with human preference. For all VLM-based metrics (CS, CT, and IS), we report the average score across three VLMs using identical system prompts: Gemini-3-Pro (Google DeepMind), GPT-5.2 (OpenAI), and Claude Sonnet-4.6 (Anthropic). Gemini-3-Pro receives native video input, while GPT-5.2 and Claude Sonnet-4.6 receive 32 uniformly sampled frames. System prompts for CS and CT are provided in the Appendix (Fig. 13).

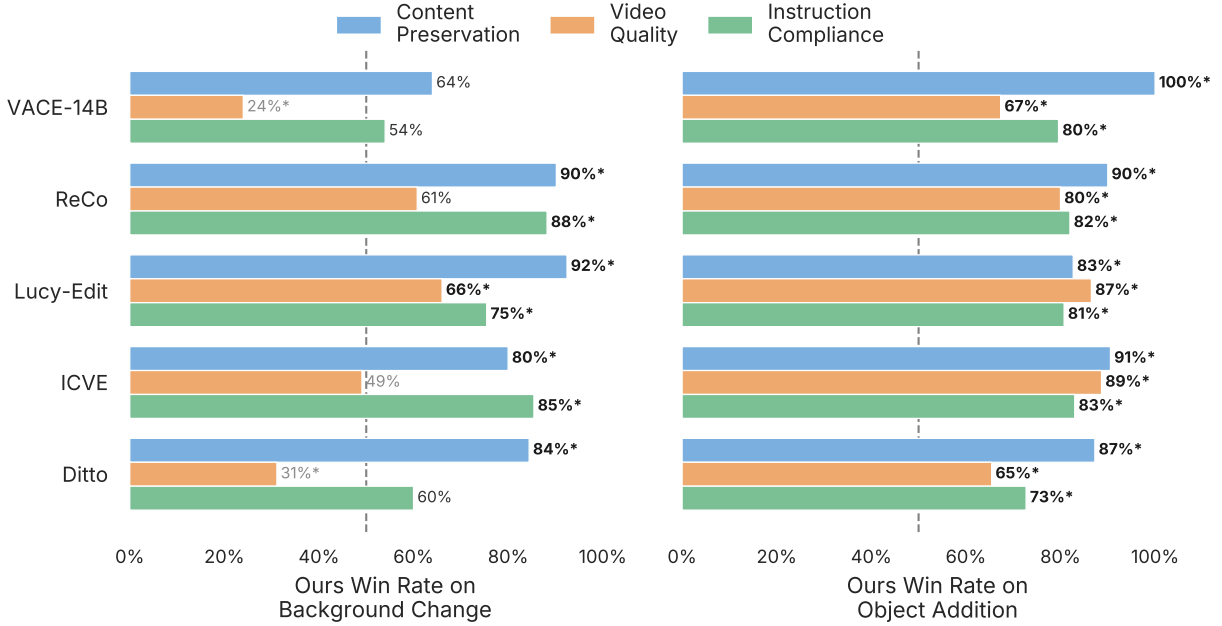


Figure 6. 2AFC user study results comparing Vera-1.3B against five baselines. Bars show our win rate across three evaluation dimensions. Bold values with * indicate statistically significant results ($p < 0.05$, binomial test) where our model is preferred.

4.1.3 Baselines

We compare against seven recent open-source video editing models spanning four categories. **General instruction-based models.** Ditto (Bai et al., 2025), Lucy-edit (Team, 2025), and ICVE (Liao et al., 2025) treat video editing as direct video-to-video translation conditioned on a text instruction. **Region-constrained editing.** ReCo (Zhang et al., 2025) introduces regional constraints during training to improve consistency in non-edited regions. **Mask-conditioned editing.** VACE (Jiang et al., 2025) (1.3B and 14B) is an all-in-one video creation and editing framework that supports both video-to-video translation and mask-conditioned editing. VideoPainter (Bian et al., 2025b) is a mask-conditioned video inpainting model. **Layered generation.** LayerFlow (Ji et al., 2025) releases separate models for different tasks; we adopt its foreground-conditioned background generation model for background change only. Note that LayerFlow is hardcoded to generate 16 frames at 720×480 resolution, so we report its performance on 16-frame sequences.

4.2 Comparison with Existing Methods

4.2.1 Quantitative Results

As reported in Table 1, Vera achieves substantially better content preservation than all baselines on both tasks. Vera-1.3B surpasses the strongest baseline by 3.5 dB PSNR on background change and by 6.3 dB on object addition, while reducing structural error ($1 - \text{SSIM}$) and perceptual distance (LPIPS) by more than half. Vera-14B extends these gains further, to 4.5 dB and 7.1 dB respectively. These improvements follow directly from the layered design, which preserves untouched content by construction, combined with the model’s accurate alpha matte prediction.

Traditional video quality metrics (OC, TF) are nearly saturated across all methods and provide little discrimination. The VLM-judged composition metrics reveal a more nuanced picture. On background change, Vera ranks second among all methods on composition quality. VACE achieves the highest CS, CT, and IS scores. This is expected given that VACE is trained on orders of magnitude more data than Vera. On object addition, Vera leads on both composition quality (CS, CT) and instruction compliance (IS) despite data disadvantages. Vera also achieves the highest IS on object addition, indicating that the layered design does not compromise edit faithfulness.

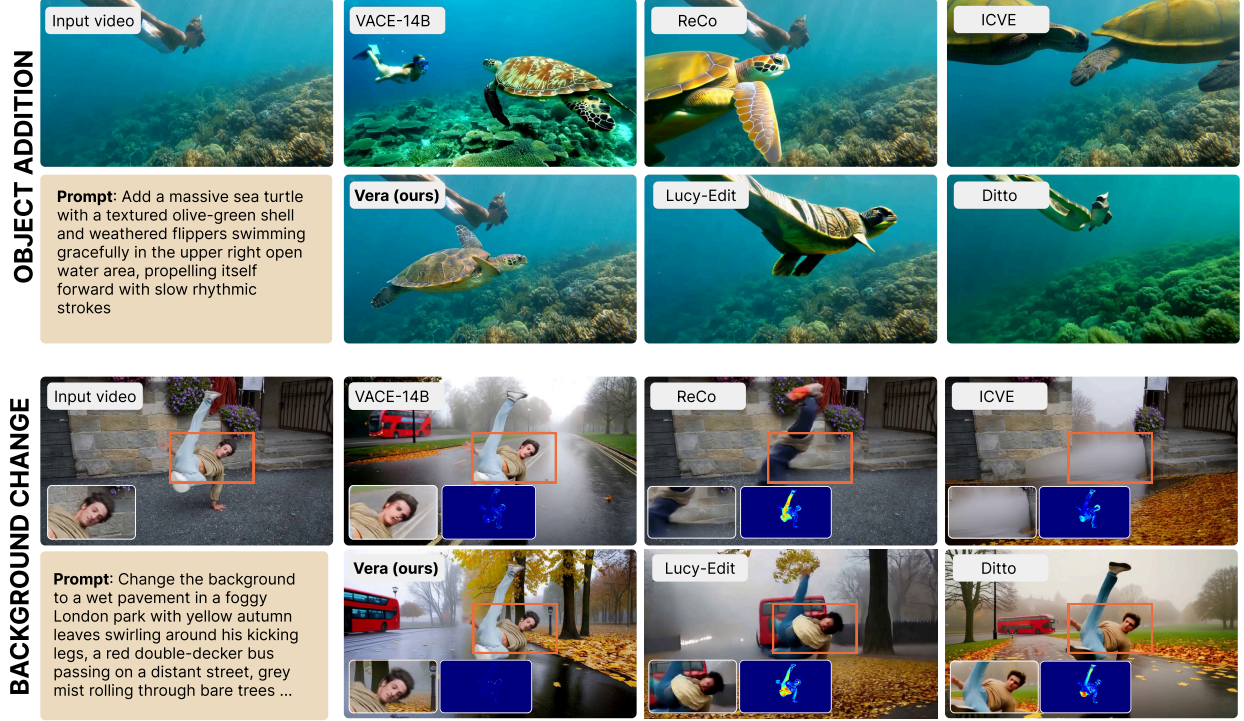


Figure 7. Qualitative comparison with existing video editing methods on background change (top) and object addition (bottom). For the background change example, each method includes sub-panels showing a zoom-in view (left) and a difference heat map over the preserved content region (right). Compared to end-to-end baselines that regenerate the entire video and introduce unintended changes to unedited regions, Vera preserves the original content best while maintaining high-quality edits.

4.2.2 User Study

To complement the automatic metrics, we conduct a human preference study using a standard two-alternative forced choice (2AFC) protocol. For each trial, annotators are shown the source video, the editing instruction, and two edited videos (Vera-1.3B vs. a randomly sampled baseline) displayed side by side with randomized left/right placement. Annotators answer three forced-choice questions, one per evaluation dimension: (1) *content preservation*: which video better preserves the original content that should remain unchanged; (2) *video quality*: which video has better overall visual quality and temporal consistency; and (3) *instruction compliance*: which video more faithfully follows the editing instruction. We recruit 19 annotators, each assigned 32 pairwise trials. We exclude 2 annotators who returned fewer than 10 preference answers, and account for a small number of incomplete trials among the remaining annotators, yielding 513 valid trials in total (see Appendix A.3 for details).

As shown in Fig. 6, Vera-1.3B is preferred over all baselines on content preservation and instruction compliance. Video quality win rates are broadly consistent with the automatic metrics, with the exception that Vera-1.3B is less preferred than VACE-14B and Ditto on background change video quality. These results are achieved with only 486K training frames, orders of magnitude less data than the baselines, suggesting that our model design is a major driver of the performance gains.

4.2.3 Qualitative Comparisons

Qualitative comparisons shown in Fig. 7 highlight two systematic failure modes in baseline methods that the quantitative metrics capture only partially. In background change, V2V baselines introduce face distortion and body morphing in regions that should remain untouched. In object addition, several baselines erroneously merge the inserted object with an existing foreground subject (e.g., fusing the sea turtle with the swimmer) or bleed its attributes (the olive-green color) into the surrounding background. Vera avoids both failure modes: the alpha matte confines all generated content to the edit layer, leaving preserved regions identical to the

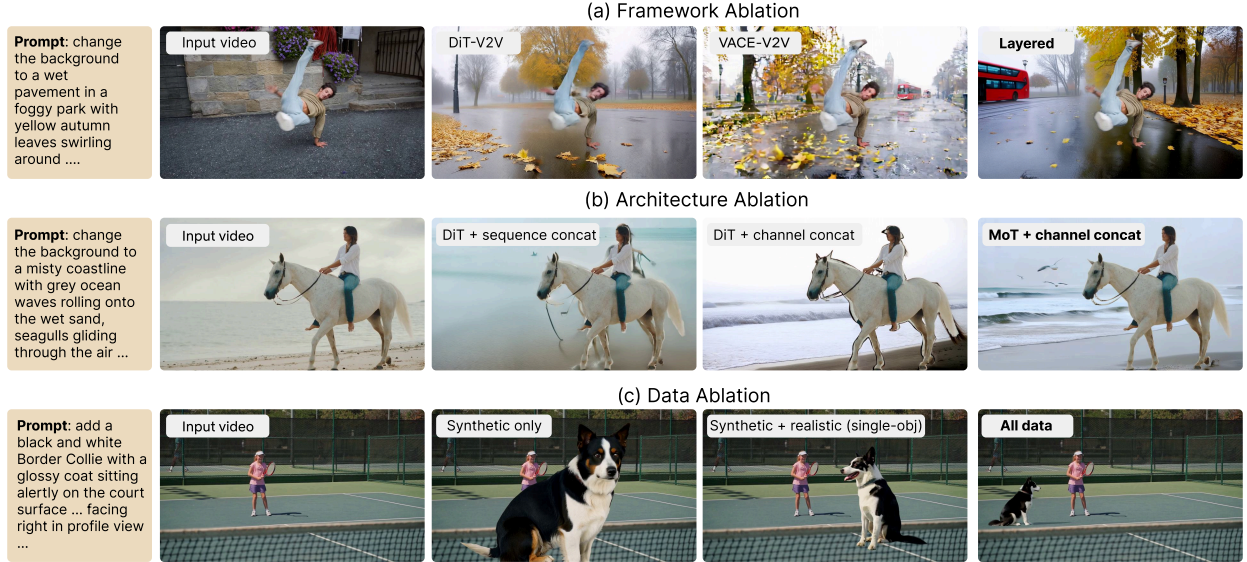


Figure 8. Qualitative examples from the ablation studies. Each row demonstrates the qualitative impact of a single design choice or training data variation while keeping all other variables fixed. **(a):** Layered editing paradigm (Vera) vs. standard video-to-video (V2V) architectures; zoom in to view the dancer’s face. **(b):** Architecture choices within the layered framework, varying DiT design (Dense DiT vs. MoT) and input video conditioning. **(c):** Cumulative effect of each training data source. **Bold** column headers indicate the choices adopted in the final Vera model.

input video.

4.3 Ablation Study

Unless otherwise noted, all ablation experiments use the Vera-1.3B model evaluated at step 6,000. VLM-based metrics (CS, CT, IS) are averaged over three VLMs, consistent with Table 1.

4.3.1 Video-to-video vs Layered Editing

Table 2 compares our layered editing paradigm against two standard video-to-video (V2V) architectures trained on the same data (architectural differences illustrated in Fig. 3). Vera achieves substantially better content preservation: +2.8 dB PSNR on background change and +3.7–4.4 dB on object addition over both V2V baselines. Crucially, video quality and instruction compliance metrics (OC, TF, CS, CT, IS) are broadly comparable across all three methods, confirming that the layered design improves preservation without sacrificing generation quality. The qualitative comparison in the first row of Fig. 8 further illustrates this: V2V baselines distort regions that should remain unchanged, while Vera preserves them exactly by construction.

4.3.2 Architecture ablation

Table 3 ablates two orthogonal design axes within the layered framework. The first axis is the *DiT design*: a single DiT that concatenates all layer tokens (edit, alpha, composite) into one sequence versus an MoT architecture with separate DiTs per layer and joint self-attention. The second axis is *input video conditioning*: channel concatenation versus sequence concatenation. For channel concatenation, we use separate patch embedding layers for the composite layer video and the input video. The composite layer’s embedding is initialized from pretrained weights; the input video embedding is either zero-initialized (*channel-zero*) or initialized by copying the pretrained weights (*channel-copy*). The MoT design is clearly superior to a single dense DiT across all three dimensions. The second row of Fig. 8 further reveals that MoT greatly improves alignment between the edit layer and the alpha matte, leading to better compositing. We also find that sequence concatenation yields better results on background change while channel concatenation is better for object addition. Since sequence concatenation incurs substantially higher FLOPs, we adopt channel concatenation for our main experiments.

Table 2. Layered editing (Vera) vs. standard video-to-video architectures. All models are trained on the same data and evaluated at step 6,000. VLM-based metrics (CS, CT, IS) are averaged over three VLMs. Best results in **bold**, second-best underlined.

Design	Content Preservation			Video Quality				Instruction Compliance	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	OC $\uparrow$	TF $\uparrow$	CS $\uparrow$	CT $\uparrow$	OSC $\uparrow$	IS $\uparrow$
<i>Background Change</i>									
Wan2.1-V2V	31.0	0.979	0.026	0.96	0.97	3.14	3.09	0.243	4.35
VACE-V2V	<u>31.0</u>	<u>0.980</u>	<u>0.025</u>	0.96	0.97	3.07	3.05	<u>0.242</u>	<u>4.34</u>
Vera (ours)	33.8	0.990	0.012	0.96	0.97	3.24	3.24	<u>0.242</u>	4.33
<i>Object Addition</i>									
Wan2.1-V2V	<u>21.1</u>	<u>0.861</u>	0.210	0.98	0.98	3.29	3.26	<u>0.243</u>	3.90
VACE-V2V	20.4	<u>0.861</u>	<u>0.209</u>	0.98	0.98	3.54	3.53	0.245	4.09
Vera (ours)	24.8	0.941	0.084	0.98	0.98	<u>3.46</u>	<u>3.48</u>	<u>0.243</u>	<u>3.97</u>

Table 3. Within-framework ablation on architecture choices for layered generation. We vary two axes: (1) DiT design – a single DiT with all layer tokens concatenated in one sequence vs. an MoT architecture with separate DiTs per layer and joint self-attention; and (2) input video conditioning – channel concatenation (with zero or copy initialization of the input video patch embedding) vs. sequence concatenation. All models evaluated at step 6,000. VLM-based metrics (CS, CT, IS) are averaged over three VLMs. Best results in **bold**, second-best underlined.

Architecture	Concat dim	Content Preservation			Video Quality				Instruction Compliance	
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	OC $\uparrow$	TF $\uparrow$	CS $\uparrow$	CT $\uparrow$	OSC $\uparrow$	IS $\uparrow$
<i>Background Change</i>										
Dense DiT	Channel	31.6	0.987	0.015	0.96	0.977	2.49	2.57	0.240	3.91
Dense DiT	Sequence	32.4	0.989	0.014	0.95	<u>0.971</u>	2.14	2.26	0.229	3.06
MoT	Sequence	34.4	0.991	0.011	0.96	0.969	3.30	3.27	0.243	4.38
MoT	Channel-copy	33.8	<u>0.990</u>	<u>0.012</u>	0.96	0.966	3.20	3.16	<u>0.242</u>	4.32
MoT	Channel-zero	<u>33.8</u>	<u>0.990</u>	<u>0.012</u>	0.96	<u>0.971</u>	<u>3.24</u>	<u>3.24</u>	<u>0.242</u>	<u>4.33</u>
<i>Object Addition</i>										
Dense DiT	Channel	21.7	0.806	0.269	0.96	0.976	1.61	1.79	0.231	2.17
Dense DiT	Sequence	20.3	0.767	0.343	0.95	<u>0.978</u>	1.32	1.46	0.223	1.75
MoT	Sequence	23.5	0.920	0.103	0.98	0.979	3.40	3.47	<u>0.243</u>	3.94
MoT	Channel-copy	<u>23.9</u>	<u>0.937</u>	<u>0.090</u>	0.98	0.979	3.49	3.50	0.244	4.01
MoT	Channel-zero	24.8	0.941	0.084	0.98	<u>0.978</u>	<u>3.46</u>	<u>3.48</u>	<u>0.243</u>	<u>3.97</u>

4.3.3 Per-layer learning rate

Table 5 investigates the effect of assigning different learning rates to the three DiT branches using Vera-1.3B. Since the edit layer, alpha matte, and composite video have substantially different distributions, each branch may benefit from a tailored learning rate. We fix a base learning rate of 10^{-5} and express each branch’s rate as a multiple of this base. Starting from a uniform rate ($1\times$ for all branches), increasing the alpha branch to $10\times$ (Config A) notably improves content preservation on background change (+2.5 dB PSNR) and instruction compliance on object addition (IS: 3.00 $\rightarrow$ 4.00). Further increasing the composite branch to $10\times$ (Config B) yields substantial gains on object addition content preservation (+3.2 dB PSNR over Config A). Finally, reducing the edit branch to $0.1\times$ (Vera’s default) yields comparable overall performance to Config B, with no clear winner between the two. We adopt the $0.1\times$ edit learning rate and fix it throughout all other experiments. Overall, the results indicate that the alpha and composite branches benefit from faster adaptation relative to the edit branch.

Table 4. Data ablation: cumulative effect of each training data source. *Synthetic* denotes VideoMatte240K (Lin et al., 2021) composites; + *Single-obj* adds realistic videos with a single prominent object; + *Multi-obj* further adds realistic videos with multiple objects. All models evaluated at step 6,000. VLM-based metrics (CS, CT, IS) are averaged over three VLMs. Best results in **bold**, second-best underlined.

Training data	Content Preservation			Video Quality				Instruction Compliance	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	OC $\uparrow$	TF $\uparrow$	CS $\uparrow$	CT $\uparrow$	OSC $\uparrow$	IS $\uparrow$
<i>Background Change</i>									
Synthetic only	<u>33.8</u>	<u>0.990</u>	<u>0.012</u>	0.96	0.977	3.21	<u>3.28</u>	0.241	4.28
+ Realistic (single-obj)	35.6	0.993	0.010	0.96	<u>0.975</u>	3.26	3.32	0.242	4.39
+ Realistic (multi-obj)	<u>33.8</u>	<u>0.990</u>	<u>0.012</u>	0.96	0.971	<u>3.24</u>	3.24	0.242	<u>4.33</u>
<i>Object Addition</i>									
Synthetic only	18.4	<u>0.823</u>	0.199	0.97	0.976	2.57	2.81	0.240	3.33
+ Realistic (single-obj)	<u>19.1</u>	0.817	<u>0.198</u>	0.98	0.978	<u>3.17</u>	<u>3.27</u>	<u>0.241</u>	<u>3.79</u>
+ Realistic (multi-obj)	24.8	0.941	0.084	0.98	0.978	3.46	3.48	0.243	3.97

Table 5. Per-layer learning rate ablation. Each DiT branch is assigned a learning rate as a multiple of the base rate 10^{-5} . Config-A increases alpha branch base learning rate by 10 $\times$. Config-B further increases composite branch learning rate by 10 $\times$. The highlighted row is Vera’s default configuration. VLM-based metrics (CS, CT, IS) are averaged over three VLMs.

Config	Learning rate			Content Preservation			Video Quality				Instruction Compliance	
	Edit	Alpha	Comp.	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	OC $\uparrow$	TF $\uparrow$	CS $\uparrow$	CT $\uparrow$	OSC $\uparrow$	IS $\uparrow$
<i>Background Change</i>												
Uniform	1x	1x	1x	32.2	0.989	0.014	0.96	0.97	2.98	3.06	0.242	4.31
Config-A	1x	10x	1x	34.7	0.991	0.011	0.96	0.97	3.08	3.13	0.241	4.34
Config-B	1x	10x	10x	33.6	0.990	0.012	0.96	0.97	3.29	3.29	0.243	4.37
Vera	0.1x	10x	10x	33.8	0.990	0.012	0.96	0.97	3.24	3.24	0.242	4.33
<i>Object Addition</i>												
Uniform	1x	1x	1x	23.1	0.871	0.160	0.98	0.98	2.30	2.45	0.237	3.00
Config-A	1x	10x	1x	22.8	0.929	0.102	0.98	0.98	3.29	3.41	0.243	4.00
Config-B	1x	10x	10x	26.0	0.956	0.072	0.98	0.98	3.36	3.48	0.243	4.00
Vera	0.1x	10x	10x	24.8	0.941	0.084	0.98	0.98	3.46	3.48	0.243	3.97

4.3.4 Data ablation

We present quantitative and qualitative study on the contribution of training data source in Table 4 and the third row of Fig. 8, respectively. The *Synthetic* layered data provides accurate alpha supervision, including fine structures such as hair, but are limited to centered human subjects with static cameras. However, as this subset lacks diversity in object scale, camera motion, and scene layout, the model receives no supervision for matching these properties and struggles to generalize to in-the-wild videos. Adding *realistic single-object* data introduces diverse, dynamic scenes that substantially boost background change preservation (PSNR: 33.8 $\rightarrow$ 35.6 dB) and improve composition quality (CS, CT) and instruction compliance (IS) on both tasks. The third dataset targets object addition with complex, multi-object scenes and interactive effects; it yields a large jump in object addition preservation (PSNR: 19.1 $\rightarrow$ 24.8 dB) with some regression on background change, suggesting that balancing the data mixture across tasks remains an important direction.

The dog example in Figure 8 nicely reflects these improvements, from having no interactions and occupying at the very front of the entire video scene with synthetic-only data, to slow gaining realistic effects such as shadows and finally becoming the right size when all data are utilized.

5 Limitations and Future Work.

The use of joint self-attention over all layer tokens makes Vera’s inference more expensive than standard V2V models; future work can apply fast sampling techniques for video diffusion models (Huang et al., 2025; Nie et al., 2026) to reduce the inference cost. On the model capability side, Vera-1.3B does not always generate background with camera motion consistent with the input video. Vera-14B largely resolves this, and we observe consistent improvement with data scaling. For object addition, the model occasionally misinterprets spatial references in the prompt as objects to generate, a common issue in instruction-based video editing models. This likely stems from the limited language understanding of the T5 text encoder (Raffel et al., 2020) and the cross-attention text conditioning in the Wan DiT; a stronger language backbone with dedicated text-modality parameters (Batifol et al., 2025) would mitigate this. Vera also cannot yet generate complex visual effects such as lightning or smoke, as these fall outside the current training distribution. Finally, the predicted alpha matte can exhibit boundary misalignment in some hard cases. However, because Vera separates edits into explicit layers, the alpha can be refined post-hoc without regenerating the edit, naturally fitting standard VFX workflows.

6 Conclusion

We introduce Vera, a layered diffusion model that separates what to generate from what to preserve. The design of Vera emphasizes both the modeling and data: with careful ablations, we identify the key architecture and data choices that enable a strong layered diffusion model with both faithful preservation and high-quality composition. The layered output format itself offers practical value: producing separable RGBA assets for compositing aligns directly with professional VFX workflows, where editors need to iteratively refine individual elements. We believe the principle of generating edits as separate RGBA layers extends naturally beyond the two tasks studied here and can serve as a foundation for more controllable video editing systems.

References

- Mixkit. Website. URL <https://mixkit.co/>. Accessed: 2026-02-16.
- Pexels. Website. URL <https://www.pexels.com/>. Accessed: 2026-02-16.
- Anthropic. Claude sonnet 4.6. <https://www.anthropic.com/>.
- Bai, Q., Wang, Q., Ouyang, H., Yu, Y., Wang, H., Wang, W., Cheng, K. L., Ma, S., Zeng, Y., Liu, Z., et al. Scaling instruction-based video editing with a high-quality synthetic dataset. *arXiv preprint arXiv:2510.15742*, 2025.
- Batifol, S., Blattmann, A., Boesel, F., Consul, S., Diagne, C., Dockhorn, T., English, J., English, Z., Esser, P., Kulal, S., et al. Flux. 1 kontext: Flow matching for in-context image generation and editing in latent space. *arXiv e-prints*, pp. arXiv–2506, 2025.
- Bian, Y., Chen, X., Li, Z., Zhi, T., Sang, S., Luo, L., and Xu, Q. Video-as-prompt: Unified semantic control for video generation. *arXiv preprint arXiv:2510.20888*, 2025a.
- Bian, Y., Zhang, Z., Ju, X., Cao, M., Xie, L., Shan, Y., and Xu, Q. Videopainter: Any-length video inpainting and editing with plug-and-play context control. *arXiv preprint arXiv:2503.05639*, 2025b.
- Caelles, S., Pont-Tuset, J., Perazzi, F., Montes, A., Maninis, K.-K., and Van Gool, L. The 2019 davis challenge on vos: Unsupervised multi-object segmentation. *arXiv*, 2019.
- Carion, N., Gustafson, L., Hu, Y.-T., Debnath, S., Hu, R., Suris, D., Ryali, C., Alwala, K. V., Khedr, H., Huang, A., et al. Sam 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.
- Chen, Y., Zhang, J., Hu, T., Zeng, Y., Xue, Z., He, Q., Wang, C., Liu, Y., Hu, X., and Yan, S. Ivebench: Modern benchmark suite for instruction-guided video editing assessment. In *The Fourteenth International Conference on Learning Representations*, 2026.
- Dalva, Y., Li, Y., Liu, Q., Zhao, N., Zhang, J., Lin, Z., and Yanardag, P. Layerfusion: Harmonized multi-layer text-to-image generation with generative priors. *arXiv preprint arXiv:2412.04460*, 2024.

- Dong, H., Wang, W., Li, C., Lyu, J., and Lin, D. Video generation with stable transparency via shiftable rgb-a distribution learner. *arXiv preprint arXiv:2509.24979*, 2025a.
- Dong, H., Wang, W., Li, C., Lyu, J., and Lin, D. Video generation with stable transparency via shiftable rgb-a distribution learner, 2025b. URL <https://arxiv.org/abs/2509.24979>.
- Esser, P., Chiu, J., Atighehchian, P., Granskog, J., and Germanidis, A. Structure and content-guided video synthesis with diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 7346–7356, 2023.
- Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024.
- Fu, F., Huang, M., Zhang, L., and Mao, Z. Layeredit: Disentangled multi-object editing via conflict-aware multi-layer learning. *arXiv preprint arXiv:2511.08251*, 2025.
- Geyer, M., Bar-Tal, O., Bagon, S., and Dekel, T. Tokenflow: Consistent diffusion features for consistent video editing. In *The Twelfth International Conference on Learning Representations*.
- Google. Veo 3.1 ingredients to video: More consistency, creativity and control. <https://blog.google/innovation-and-ai/technology/ai/veo-3-1-ingredients-to-video/>, 2026.
- Google DeepMind. Gemini 3 pro. <https://deepmind.google/technologies/gemini/>.
- HaCohen, Y., Chiprut, N., Brazowski, B., Shalem, D., Moshe, D., Richardson, E., Levin, E., Shiran, G., Zabari, N., Gordon, O., Panet, P., Weissbuch, S., Kulikov, V., Bitterman, Y., Melumian, Z., and Bibi, O. Ltx-video: Realtime video latent diffusion. *arXiv preprint arXiv:2501.00103*, 2024.
- Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- Hu, J., Zhong, T., Wang, X., Jiang, B., Tian, X., Yang, F., Wan, P., and Zhang, D. Vivid-10m: A dataset and baseline for versatile and interactive video local editing. *arXiv preprint arXiv:2411.15260*, 2024.
- Huang, X., Li, Z., He, G., Zhou, M., and Shechtman, E. Self forcing: Bridging the train-test gap in autoregressive video diffusion. *arXiv preprint arXiv:2506.08009*, 2025.
- Huang, Z., He, Y., Yu, J., Zhang, F., Si, C., Jiang, Y., Zhang, Y., Wu, T., Jin, Q., Chanpaisit, N., Wang, Y., Chen, X., Wang, L., Lin, D., Qiao, Y., and Liu, Z. VBench: Comprehensive benchmark suite for video generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- Ji, S., Luo, H., Chen, X., Tu, Y., Wang, Y., and Zhao, H. Layerflow: A unified model for layer-aware video generation. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pp. 1–10, 2025.
- Jia, B., Shao, Z., Li, X., Fan, J., Chen, R., and Wang, B. Cococo: Improving text-guided video inpainting for better consistency, controllability and compatibility. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- Jiang, Z., Han, Z., Mao, C., Zhang, J., Pan, Y., and Liu, Y. Vace: All-in-one video creation and editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 17191–17202, 2025.
- Kong, W., Tian, Q., Zhang, Z., Min, R., Dai, Z., Zhou, J., Xiong, J., Li, X., Wu, B., Zhang, J., et al. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*, 2024.
- Lee, Y.-C., Lu, E., Rumbley, S., Geyer, M., Huang, J.-B., Dekel, T., and Cole, F. Generative omnimatte: Learning to decompose video into layers. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 12522–12532, 2025.

- Li, B., Zhang, Y., Wang, Q., Ma, L., Shi, X., Wang, X., Wan, P., Yin, Z., Zhuge, Y., Lu, H., et al. Vfxmaster: Unlocking dynamic visual effect generation via in-context learning. *arXiv preprint arXiv:2510.25772*, 2025.
- Liang, W., YU, L., Luo, L., Iyer, S., Dong, N., Zhou, C., Ghosh, G., Lewis, M., tau Yih, W., Zettlemoyer, L., and Lin, X. V. Mixture-of-transformers: A sparse and scalable architecture for multi-modal foundation models. *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856. URL <https://openreview.net/forum?id=Nu6N69i8SB>.
- Liao, X., Zeng, X., Song, Z., Fu, Z., Yu, G., and Lin, G. In-context learning with unpaired clips for instruction-based video editing. *arXiv preprint arXiv:2510.14648*, 2025.
- Lim, S., Oh, S. W., Huang, J., Yoon, H., Kim, S., and Lee, J.-Y. Videomama: Mask-guided video matting via generative prior. *arXiv preprint arXiv:2601.14255*, 2026.
- Lin, S., Ryabtsev, A., Sengupta, S., Curless, B. L., Seitz, S. M., and Kemelmacher-Shlizerman, I. Real-time high-resolution background matting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8762–8771, 2021.
- Lipman, Y., Chen, R. T., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*.
- Litman, Y., Liu, S., Seyb, D., Milef, N., Zhou, Y., Marshall, C., Tulsiani, S., and Leak, C. Editctrl: Disentangled local and global control for real-time generative video editing. *arXiv preprint arXiv:2602.15031*, 2026.
- Liu, X., Gong, C., et al. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *The Eleventh International Conference on Learning Representations*.
- Liu, X., Gong, C., and Liu, Q. Flow straight and fast: Learning to generate and transfer data with rectified flow. *arXiv preprint arXiv:2209.03003*, 2022.
- Nie, W., Berner, J., Ma, N., Liu, C., Xie, S., and Vahdat, A. Transition matching distillation for fast video generation. *arXiv preprint arXiv:2601.09881*, 2026.
- OpenAI. GPT-5.2. <https://openai.com/>.
- OpenAI. Sora 2 is here, September 2025. URL <https://openai.com/index/sora-2/>.
- Pont-Tuset, J., Perazzi, F., Caelles, S., Arbeláez, P., Sorkine-Hornung, A., and Van Gool, L. The 2017 davis challenge on video object segmentation. *arXiv:1704.00675*, 2017.
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., and Liu, P. J. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21 (140):1–67, 2020.
- Ravi, N., Gabeur, V., Hu, Y.-T., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., et al. Sam 2: Segment anything in images and videos. In *The Thirteenth International Conference on Learning Representations*.
- Runway. Runway aleph. <https://runwayml.com/research/introducing-runway-aleph>, 2025.
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*.
- Tang, Z., Yang, Z., Zhu, C., Zeng, M., and Bansal, M. Any-to-any generation via composable diffusion. *Advances in Neural Information Processing Systems*, 36:16083–16099, 2023.
- Team, C. Chameleon: Mixed-modal early-fusion foundation models. *arXiv preprint arXiv:2405.09818*, 2024.
- Team, D. Lucy edit: Open-weight text-guided video editing. *blog post*, 2025. URL https://d2drjpuinn461b.cloudfront.net/Lucy_Edit_High_Fidelity_Text_Guided_Video_Editing.pdf.

- Wan, T., Wang, A., Ai, B., Wen, B., Mao, C., Xie, C.-W., Chen, D., Yu, F., Zhao, H., Yang, J., et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025a.
- Wan, T., Wang, A., Ai, B., Wen, B., Mao, C., Xie, C.-W., Chen, D., Yu, F., Zhao, H., Yang, J., et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025b.
- Wang, L., Li, Y., Chen, Z., Wang, J.-H., Zhang, Z., Zhang, H., Lin, Z., and Chen, Y.-C. Transpoxler: Advancing text-to-video generation with transparency. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 18229–18239, 2025.
- Yang, J., Liu, Q., Li, Y., Kim, S. Y., Pakhomov, D., Ren, M., Zhang, J., Lin, Z., Xie, C., and Zhou, Y. Generative image layer decomposition with visual effects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 7643–7653, 2025.
- Yang, Z., Teng, J., Zheng, W., Ding, M., Huang, S., Xu, J., Yang, Y., Hong, W., Zhang, X., Feng, G., et al. Cogvideox: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024.
- Yin, S., Zhang, Z., Tang, Z., Gao, K., Xu, X., Yan, K., Li, J., Chen, Y., Chen, Y., Shum, H.-Y., et al. Qwen-image-layered: Towards inherent editability via layer decomposition. *arXiv preprint arXiv:2512.15603*, 2025.
- Zhang, H., Xing, Z., Xu, Q., Jiang, X., Zhu, H., Xu, C., and Zha, Z.-J. Region-constraint in-context generation for instructional video editing. *arXiv preprint arXiv:2512.17650*, 2025.
- Zhang, L. and Agrawala, M. Transparent image layer diffusion using latent transparency. *ACM Trans. Graph.*, 43(4), July 2024. ISSN 0730-0301. doi: 10.1145/3658150. URL <https://doi.org/10.1145/3658150>.
- Zhou, Y., Bu, J., Ling, P., Zhang, P., Wu, T., Huang, Q., Li, J., Dong, X., Zang, Y., Cao, Y., et al. Light-a-video: Training-free video relighting via progressive light fusion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 13315–13325, 2025.
- Zi, B., Ruan, P., Chen, M., Qi, X., Hao, S., Zhao, S., Huang, Y., Liang, B., Xiao, R., and Wong, K.-F. Señorita-2m: A high-quality instruction-based dataset for general video editing by video specialists. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Zi, B., Peng, W., Qi, X., Wang, J., Zhao, S., Xiao, R., and Wong, K.-F. Minimax-remover: Taming bad noise helps video object removal. *arXiv preprint arXiv:2505.24873*, 2025.

A Supplementary Material

A.1 Training data

A.1.1 Data pipeline overview

As described in the main paper, our layered training data is constructed from two types of video sources: the video matting dataset (Lin et al., 2021), and real-world videos collected from Pexels (pex) and Mixkit (mix). Both pipelines are multi-stage processes that combine automated tools (*e.g.*, segmentation, matting, inpainting, and VLM-based captioning with Gemini-3-Pro) with human annotators, who identify suitable objects, provide point prompts to SAM2 for segmentation, and filter out low-quality samples at each stage.

Fig. 5 illustrates the complete data construction pipelines for (a) object addition and (b) background change, showing the full processing chain starting from real internet videos. For the VideoMatte240K dataset, since it already provides high-quality alpha mattes with precise boundaries, the first two stages—video collection/filtering and matting—can be skipped, and the pipeline proceeds directly from the object removal stage onward. We describe each pipeline below. Table 8 lists all data sources, models, and tools used in the data pipelines along with their licenses.

Object addition pipeline. The object addition pipeline (Fig. 5a) consists of six stages: **(1) Video collection and filtering.** We collect raw videos from Pexels (pex) and Mixkit (mix), then filter and preprocess them to obtain a set of high-quality videos with sufficient resolution and visual diversity, covering various scenes and objects. **(2) Matting.** Human annotators identify suitable object(s) in each video and provide point prompts to SAM2 (Ravi et al.). The SAM2 segmentation masks are then refined with VideoMaMa (Lim et al., 2026) to obtain high-quality alpha mattes with accurate boundaries. **(3) Object removal.** Using the alpha mattes, we remove the selected object(s) from the video with a video object removal model (Casper-1.3B (Lee et al., 2025)) to obtain clean background videos. Since the removal model can occasionally produce artifacts, this stage includes a filtering step to discard failed samples. **(4) Omnimatte optimization.** For objects with associated visual effects (*e.g.*, shadows and reflections), we apply omnimatte optimization (Lee et al., 2025) to extract alpha mattes that capture not only the object itself but also its effects. This step is skipped for objects without associated effects. **(5) Effect refinement.** The omnimatte optimization results are often noisy. Human annotators identify regions containing actual effects (as opposed to artifacts) and use SAM2 to segment these regions. The resulting mask is used to refine the alpha by zeroing out regions beyond the mask, producing clean edit layers and alpha mattes. **(6) Captioning.** Finally, a VLM generates edit instructions describing the object addition (see Fig. 10 for the system prompt), producing the final (input video, edit instruction, edit layer, alpha matte) training tuple.

Background change pipeline. The background change pipeline (Fig. 5b) shares the first two stages (video collection and matting) with the object addition pipeline—in practice, we often reuse the alpha matte results directly. It then diverges as follows: **(3) Object removal.** As in the object addition pipeline, we remove the identified object(s) from the video using Casper-1.3B (Lee et al., 2025). However, unlike object addition, here we want the object’s visual effects (*e.g.*, shadows, reflections) to remain in the background, since the background with effects will serve as our edit layer. **(4) Effects transfer.** We transfer pixels from the original video that lie beyond the alpha matte region back into the object-removed video, so that only the object itself is removed while its associated effects are preserved in the background. The resulting video—the original background with effects intact but the object removed—becomes the edit layer. **(5) Background generation and recomposition.** We use a VLM to analyze the matted object(s) and generate scene prompts describing backgrounds that would naturally fit with them. These prompts are sent to VACE inpainting (Jiang et al., 2025) to generate a synthetic background, which is then composited with the object using the alpha matte to produce the input video. **(6) Captioning.** We first use a VLM to generate detailed captions of both the objects and the scene for the input video and the target composite using the system prompt in Fig. 11. These captions are then provided to a separate VLM call (Fig. 12) to infer the background change editing instruction.

A.2 Model details

Table 6 summarizes the architecture configurations. Vera extends the base Wan2.1 T2V DiT into a MoT with three separate DiTs—one each for the edit layer, alpha matte, and composite video—that interact through joint self-attention. The number of transformer layers is inherited from the base model (30 for the 1.3B variant

and 40 for the 14B variant). Because each DiT maintains its own set of parameters, the total parameter count scales by a factor of three relative to the base T2V model. However, per-step FLOPs increase by more than 3 $\times$ because joint self-attention operates over the combined token sequence from all three DiTs, and attention cost grows quadratically with sequence length.

Table 6. Architecture comparison between the base Wan2.1 T2V models and the corresponding Vera variants. TFLOPs is measured per denoising step.

Model	# Parameters	TFLOPs	Layers	Output Layers
Wan2.1-T2V-1.3B	1.3B	129	30	1
Vera-1.3B	3.9B	840	30	3
Wan2.1-T2V-14B	14B	832	40	1
Vera-14B	42B	4,514	40	3

Training details We train Vera with Adam, using individual base learning rates for each DiT as determined by the per-layer learning-rate ablation in Table 5. We apply a linear warmup of 200 steps. To encourage the model to generate both with and without the mask input condition, we randomly drop the mask input during training. We use a mask dropout rate of 0.3 for background change and 0.9 for object addition. The losses for the three output layers are equally weighted. Each of the five input streams—edit layer latent, alpha layer latent, composite layer latent, mask video latent, and input video latent—has its own dedicated patch embedding layer. The patch embeddings for the mask video and input video are initialized from zero, while the remaining patch embedding layers are initialized from the pre-trained T2V model.

A.3 User study

Fig. 9 shows the annotation interface used for our human preference study. As described in the main paper, we adopt a two-alternative forced choice (2AFC) protocol. Each trial presents the annotator with the source video (top left), the editing instruction (top right), and two anonymized edited videos (Video A and Video B) displayed side by side in the bottom row. The assignment of Vera and the baseline to the left or right position is randomized per trial to avoid positional bias. To aid content preservation judgments, difference heatmaps (Diff A and Diff B) are displayed alongside each video, highlighting pixel-level deviations from the source. Annotators then answer three forced-choice questions covering content preservation, video quality, and instruction compliance. We recruited 19 annotators, each assigned 32 pairwise trials, for a maximum of $19 \times 32 = 608$ trials. We exclude the responses of annotators who returned fewer than 10 preference answers, which filters out 2 annotators and leaves 17 effective annotators ($17 \times 32 = 544$ trials). Among the remaining annotators, a small number of trials were not completed due to technical interruptions (*e.g.*, network connectivity issues) or annotator abstentions, yielding 513 valid trials in total.

New User -- Click to Start

Your User ID
user-n1pb

Progress
1 / 32

Source Video

Edit Prompt
Change the background to a serene Nordic forest path in winter with snow-covered pine trees, snowflakes gently drifting in the air, breath condensing into visible vapor trails as she runs, soft blue shadows on the snow, soft cool back lighting from natural daylight. Medium shot, eye-level, tracking camera, shallow depth of field. Keep the athletic woman running in black sportswear intact.

Diff A

Video A

Video B

Diff B

Content Preservation: Which video better preserves the original content that should remain unchanged?
☐ Video A ☐ Video B

Video Quality: Which video has better overall visual quality and temporal consistency?
☐ Video A ☐ Video B

Instruction Compliance: Which video more faithfully follows the editing instructions?
☐ Video A ☐ Video B

Previous (P)

Skip (broken video)

Submit & Next (N)

Figure 9. Screenshot of the annotation interface used in our human preference study. Each user is assigned an anonymous id. For each trial, annotators view the source video and the editing instruction at the top, followed by two anonymized edited videos (Video A and B) with their corresponding difference heatmaps (Diff A and B). Annotators select their preference for each of the three evaluation dimensions via forced-choice buttons.

Table 7. Evaluation metrics grouped by dimension. Arrows indicate whether higher (↑) or lower (↓) is better.

Dimension	Metric	Description
Content Preservation	PSNR ↑	Peak signal-to-noise ratio
	SSIM ↑	Structural similarity
	LPIPS ↓	Learned perceptual similarity
Video Quality	OC ↑	Overall consistency: average of subject consistency (DINO (Huang et al., 2024)) and background consistency (CLIP (Huang et al., 2024)) across frames
	TF ↑	Temporal flickering: mean frame difference (Huang et al., 2024)
	CS ↑	Composition spatial quality: VLM-judged per-frame spatial blending, averaged over 3 VLMs (ours)
	CT ↑	Composition temporal quality: VLM-judged temporal coherence of the edit, averaged over 3 VLMs (ours)
Instruction Compliance	OSC ↑	Overall semantic consistency between edited video and target prompt via VideoCLIP-XL2 (Chen et al., 2026)
	IS ↑	Instruction satisfaction: VLM-judged accuracy on a 5-point scale, averaged over Gemini-3-Pro, GPT-5.2, and Claude Sonnet-4.6 (Chen et al., 2026)

Table 8. Licenses for datasets, annotation tools, and foundation models used in this work.

Data / Model / Tool	License
Pexels videos (pex)	Pexels Free License
Mixkit videos (mix)	Mixkit Free License
VideoMatte240K (Lin et al., 2021)	MIT License
DAVIS (Pont-Tuset et al., 2017 ; Caelles et al., 2019)	BSD 3-Clause License
VACEBench (Jiang et al., 2025)	Apache-2.0 License
SAM2 (Ravi et al.)	Apache-2.0 License
VideoMama (Lim et al., 2026)	CC BY-NC 4.0
VACE (Jiang et al., 2025)	Apache-2.0 License
Minimax-Remover (Zi et al., 2025)	CC BY-NC 4.0
Generative Omnimatte (Lee et al., 2025)	Apache-2.0 License
Wan2.1-14B (Wan et al., 2025b)	Apache-2.0 License

You are an expert video editing instructor. You will receive two videos: a "before" video (clean background without the object) and an "after" video (with the object composited in). Your task is to generate a detailed editing instruction describing the added object.

Your Task

Compare the two videos carefully and write a thorough instruction describing everything about the added subject — its appearance, motion, and placement in the scene. Provide as much visual detail as the videos justify; do not artificially shorten or summarize your description.

Prompt Formula:

Add [detailed subject description] [action/motion with specifics] [precise location/placement in the scene]. [Camera and lighting constraints].

What to Describe

Subject Appearance

- Physical attributes: size, shape, build, posture
- Colors and textures: clothing materials, fur, skin tone, surface finish
- Distinctive features: accessories, patterns, markings, facial expression
- If the subject is a person: clothing details (garments, colors, fit), hair, visible accessories
- If the subject is an animal: breed/species cues, coat color and texture, size relative to scene
- If the subject is an object: material, color, condition, style

Motion and Action

- Primary action: what the subject is doing (e.g., walking, sitting, spinning, hovering)
- Motion direction and speed: left to right, toward camera, slowly, briskly
- Body dynamics: limb positions, posture changes, secondary motion (hair swaying, clothes fluttering)
- Interaction: if the subject interacts with anything already in the scene, describe the interaction

Spatial Placement

- Position in frame: left/center/right, foreground/midground/background
- Relative scale: how large the subject appears compared to the scene
- Facing direction — include for any subject that has a distinguishable front (people, animals, vehicles, aircraft, etc.). Omit only for symmetric objects with no clear front (e.g., a ball, a box, a cup):

Term	What you see

facing the camera	Front visible (face, headlights, nose cone, etc.)
three-quarter view	Angled between front and side
profile view (left/right)	Pure side view
facing away	Rear toward the camera (back of head, tail, tailgate, etc.)
looking over the shoulder	Body facing away but head turned back (people/animals only)

- Depth placement: how far from the camera the subject appears

Lighting and Shadow Interaction

- If visible: describe any shadows cast by the added subject
- Describe how the lighting falls on the subject (lit from which side, rim-lit, silhouetted, etc.)

Camera Constraints

- Shot size as it relates to the subject: how much of the subject is visible (full body, waist up, close-up, etc.)
- Camera angle relative to the subject: eye-level, low angle, high angle, etc.
- Background focus around the subject: blurred background, sharp background, soft bokeh, etc.

Rules

DO:

- Start with "Add"
- Describe only what is clearly visible — be specific and concrete
- Include all visually relevant details about the added subject
- Always include facing direction for people and animals (e.g., facing the camera, facing away, profile view, three-quarter view). This is mandatory — do not omit it
- Include camera constraints: shot size, camera angle, and background focus
- Keep the instruction under 80 words — prioritize the most visually important details
- Describe motion over time if it changes across the video

DON'T:

- Do NOT describe the background or environment — only describe the added object and its interaction with the scene
- Do NOT reference what was in the "before" video or what the object replaces — describe the added subject as if it simply appears in the scene
- Do NOT hallucinate details that are not clearly visible in the videos
- Do NOT use vague descriptors like "nice", "interesting"

Output Format

Return a JSON object with a single "instruction" field containing the editing instruction.

Examples

Output: {"instruction": "Add a yellow taxi cab parked diagonally along the left side of the street, front bumper angled toward the camera, glossy paint reflecting the overcast sky, license plate partially visible, faint shadow cast underneath onto the wet asphalt. Wide shot, eye-level, sharp background."}

Output: {"instruction": "Add a young child in a royal-blue puffer jacket and dark jeans sitting cross-legged on the right end of the wooden bench, facing the camera, leaning slightly forward with hands on their knees, quilted texture catching overhead light with small highlights along the seams. Medium shot, slightly high angle, slightly blurred background."}

Output: {"instruction": "Add a white domestic shorthair cat walking steadily from right to left along the windowsill in the foreground, facing left in profile, tail held upright curving at the tip, each paw placed precisely on the narrow ledge, soft side-lighting from the window on the left side of its body. Medium close-up, eye-level, dreamy out-of-focus background."}

Figure 10. The system prompt used for generating edit instructions for the object addition data.

You are an expert cinematographer and video description AI. Your task is to write a concise, accurate caption for a given video, suitable for use as a prompt for AI video diffusion models.

****Input:**** You will receive a video. Watch the video and write a high-quality prompt to describe the video.

****Output Format:**** Write a high-quality prompt following this formula:

```
...
[Subject(s) description] + [action/motion] + [environment/scene] + [Constraints: lighting/color description + shot/camera description]
...
```

For subject(s), action, and environment: compose natural, flowing prose. If there are multiple subjects, describe each subject's appearance and action clearly.

For constraints (lighting + camera): use a concise list of key short phrases or terms. See examples below for the expected style.

Rules

DO:

1. ****Describe the subject(s) accurately**** - appearance, clothing, distinctive features. If multiple subjects are present, describe each one and how they relate spatially
2. ****Describe the motion with details**** - what each subject is doing, speed and intensity
3. ****Describe the environment with details**** - location, setting, background elements
4. ****Include lighting constraints**** - direction, quality, color temperature, source
5. ****Include camera constraints**** - shot size, focal length feel, angle, motion, background focus
6. ****Be specific**** - textures, colors, spatial relationships

DON'T:

1. ****Don't use vague descriptors**** - avoid "nice", "beautiful", "interesting"
2. ****Don't overcomplicate**** - one scene, one action, clear description
3. ****Don't add interpretation**** - describe what you see, not what you infer
4. ****Don't include JSON or structured formatting**** - output plain text only
5. ****Never say "depth of field"**** - instead use the background focus phrases from the vocabulary (e.g., "blurred background", "soft bokeh", "sharp background", "deep focus"). The phrase you choose must be consistent with the actual background sharpness in the video — do not contradict yourself
6. ****Always include focal length feel**** - wide-angle, normal, or telephoto

Vocabulary Reference

****Shot sizes:****

Term	Frame Coverage
extreme wide shot	Vast environment, subject very small or barely visible
wide shot	Full body + significant environment around subject
full shot	Head to toe, subject fills frame vertically
medium wide shot	Knees up ("cowboy shot")
medium shot	Waist up
medium close-up	Chest up
close-up	Face/head fills frame
extreme close-up	Single detail only (eyes, hands, object)

****Camera motion:****

Term	Description
static	No camera movement, tripod shot
pan left/right	Camera rotates horizontally on axis
tilt up/down	Camera rotates vertically on axis
dolly in/out	Camera moves toward/away from subject
tracking	Camera moves parallel to subject
crane	Camera moves up/down vertically
orbiting	Camera circles around subject
handheld	Slight shake, organic movement
zoom in/out	Lens focal length changes, no physical camera movement

Examples

****Input:**** Video of a man walking on a sidewalk

****Output:****

A man in a blue jacket walks steadily along a tree-lined sidewalk in a residential neighborhood, with cars parked along the street and leaves drifting in the breeze. Soft diffused overcast daylight with neutral color temperature, even illumination with minimal shadows. Medium shot, normal lens, eye-level, tracking right, slightly blurred background.

****Input:**** Video of a woman dancing in a studio

****Output:****

A woman in a black leotard spins gracefully across a polished wooden floor in a dance studio, arms extended and hair flowing with the motion. Warm tungsten side-lighting from the right casting long shadows across the floor. Full shot, normal lens, low angle, static camera, blurred background with soft bokeh.

****Input:**** Video of a dog running on a beach

****Output:****

A golden retriever sprints along the wet shoreline, kicking up sand and water with each stride as waves crash in the background. Golden hour back-light creates a warm rim light around the dog's fur. Medium wide shot, telephoto lens with compressed depth, eye-level, tracking left, sharp background, deep focus.

Figure 11. The system prompt used for generating video captions.

You are an expert at writing video editing instructions. Your task is to generate a concise background change instruction.

Input
You will receive two complete scene prompts, each containing:

- Subject description and motion (e.g., "A woman in a white dress dancing")
- Scene/environment description (e.g., "in an enchanted forest clearing at twilight")
- Lighting and style constraints (e.g., "Soft warm light from the left")

- Source scene: The complete prompt describing the original video
- Target scene: The complete prompt describing the desired result

Your Task
Extract the background/environment information from the TARGET scene and write an editing instruction.

The instruction should:

- Start with "Change the background to"
- Describe ONLY the target environment/scene (extracted from target prompt)
- Include relevant lighting, atmosphere, or style details from the target
- Include camera constraints if present (shot type, angle, depth of field, etc.)
- End with "Keep the [brief subject description] intact." (e.g., "Keep the woman in the white dress intact.")

Output Format
Return a JSON object with a single "instruction" field containing the editing instruction.

Examples

****Source scene:**** "A woman in a white dress dancing in an enchanted forest clearing at twilight. Magical fireflies glowing around her. Soft warm light from the left."

****Target scene:**** "A woman in a white dress dancing on a modern rooftop at sunset. City skyline glowing in the background. Warm golden hour light from the right."

****Output:**** "Change the background to a modern rooftop at sunset with a glowing city skyline, warm golden hour lighting from the right. Keep the woman in the white dress intact."

****Source scene:**** "A child running through a sunny meadow with wildflowers. Bright natural daylight, blue sky with fluffy clouds. Wide shot, shallow depth of field."

****Target scene:**** "A child running along a sandy beach at dawn. Gentle waves in the background, soft pink and orange sunrise colors. Medium shot, deep focus."

****Output:**** "Change the background to a sandy beach at dawn with gentle waves, soft pink and orange sunrise lighting. Medium shot, deep focus. Keep the child intact."

Figure 12. The system prompt used for generating edit instructions for the background change data.

You are a video editing quality assessment expert. You are given an original video, an edited video, and the editing instruction: '{edit_prompt}'.

By comparing the original and edited videos, assess the visual quality of the edit on two dimensions, each on a scale from 1 to 5. Consider both the quality of the edited content AND the preservation of regions that should remain unchanged (infer which regions from the editing instruction).

****Spatial Quality**** (per-frame visual quality):

- Edited regions: Do the new or modified elements look natural and realistic? Are they free of AI artifacts (distorted faces/hands, unnatural textures, blurriness)?
- Edited regions: Is the lighting, color, and perspective of edited content consistent with the rest of the scene?
- Edited regions: Do the edits blend seamlessly into the scene without visible seams, halos, or boundary artifacts?
- Preserved regions: Has the visual quality (sharpness, detail, color) of non-edited areas been maintained compared to the original?
- Overall: Does the edited video look like a coherent, realistic scene?

Rating scale: 1 = severe quality issues (obvious artifacts, distortions, or degradation), 2 = noticeable quality problems, 3 = acceptable with minor issues, 4 = good quality with minimal issues, 5 = excellent, indistinguishable from a real video

****Temporal Quality**** (across-frame quality over time):

- Edited regions: Does the edited content maintain consistent appearance across frames (no morphing, flickering, or sudden changes)?
- Edited regions: Is the motion of edited content natural and physically plausible?
- Edited regions: Does the edited content move consistently with the scene under camera motion (no sliding, drifting, or floating)?
- Preserved regions: Has the temporal quality of non-edited areas been maintained (no introduced flickering, jitter, or instability)?
- Overall: Does the video flow naturally without temporal discontinuities?

Rating scale: 1 = severe temporal issues (obvious flickering, sliding, or instability), 2 = noticeable temporal problems, 3 = acceptable with minor issues, 4 = good temporal consistency, 5 = excellent, seamless temporal quality

Respond with ONLY a JSON object: `{{"spatial": <1-5>, "temporal": <1-5>, "explanation": "<brief reasoning>"}}`

Figure 13. The system prompt for VLM-judged composition spatial quality (CS) and composition temporal quality (CT).